Lab Report - Lab Unit 1

System Sketches

Software and Process Notations

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Objective/Physical-Based

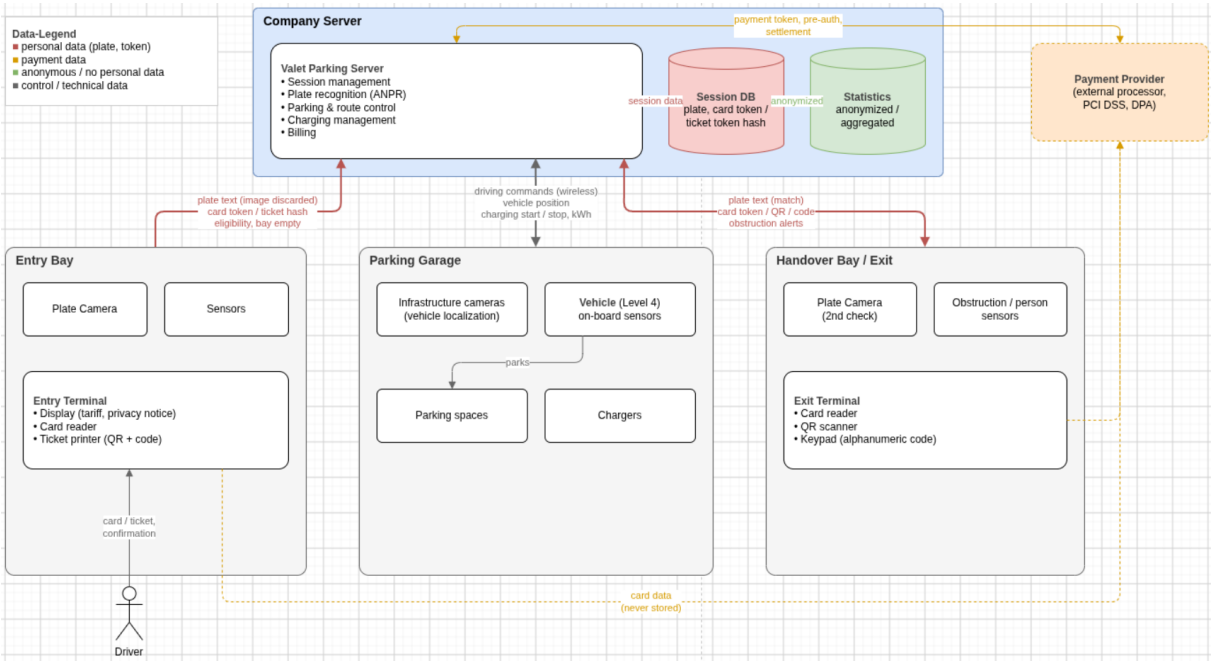


Figure 1: Objective/physical-based view: components of the system and their interactions

Process-Based

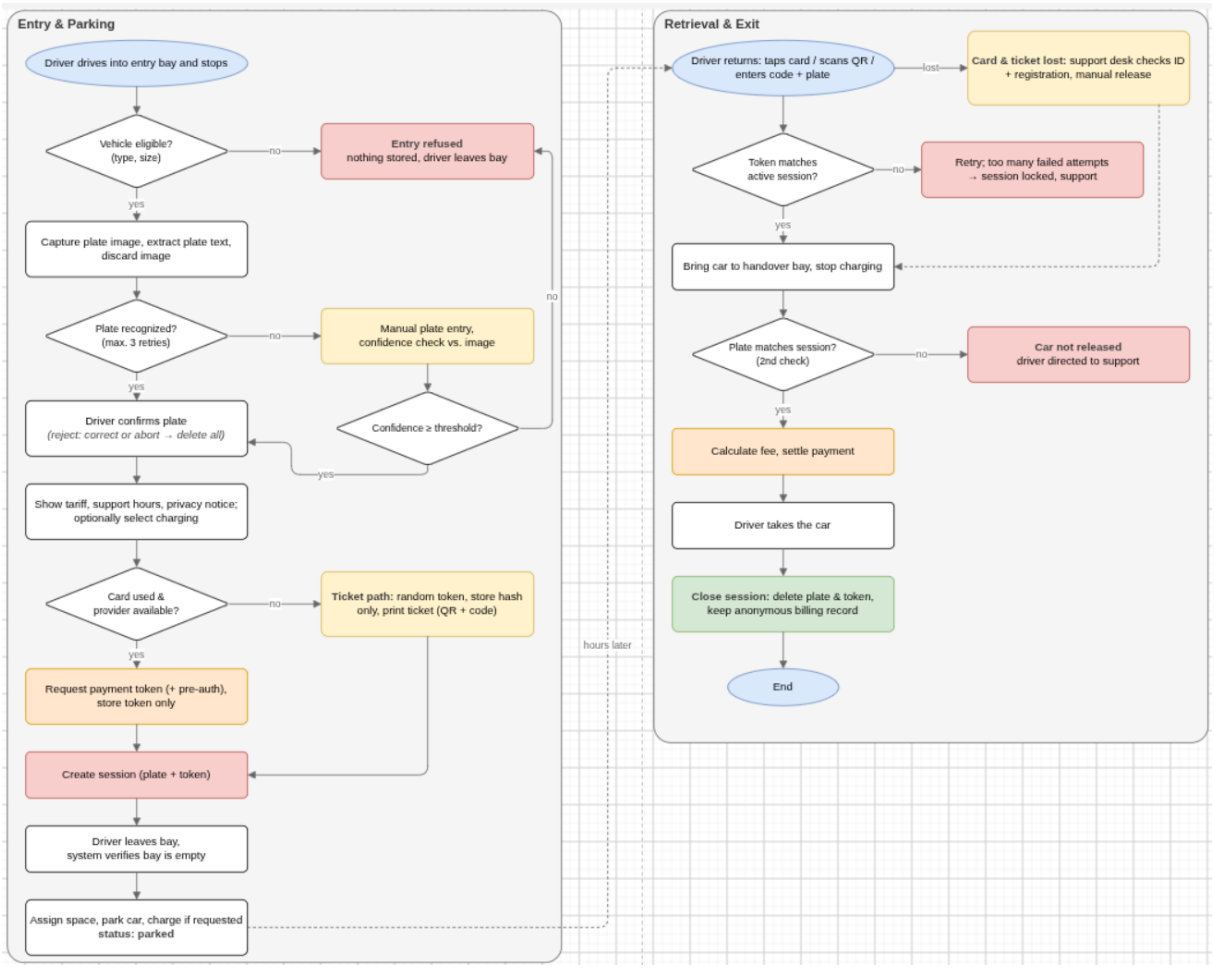


Figure 2: Process-based view: sequence of actions incl. alternate courses and exceptions

Time-Based

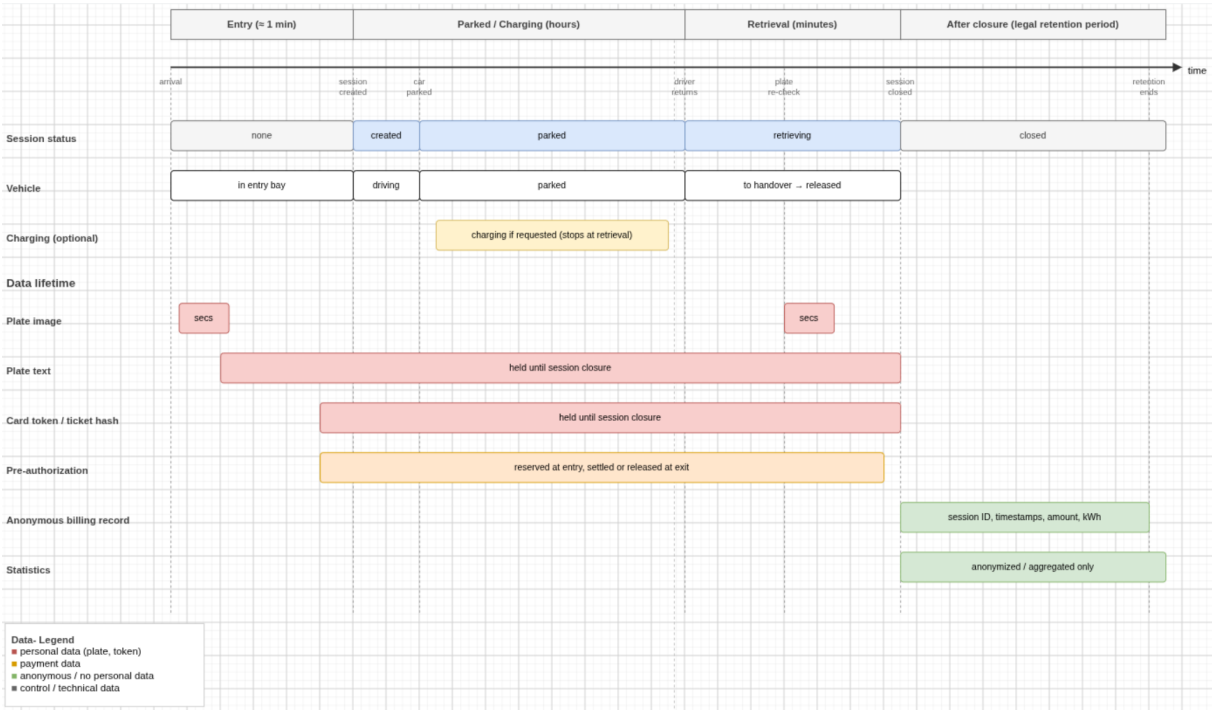


Figure 3: Time-based view: system behaviour and data lifetime over one parking session

Statement on the Use of AI

In preparing this lab report, Claude Code (Anthropic, Claude Opus 5.5, token based billing) was used as a generative AI tool exclusively to assist with the following tasks:

- creating and designing the LaTeX layout,
- drafting the three system sketches (draw.io),
- checking spelling and grammar,
- research.

The LaTeX template of this report, including this statement, was created with the assistance of the AI tool and reviewed and adapted by the lab members. First drafts of the three system sketches were generated by the AI tool based on the use case description written by the lab members; the lab members subsequently reviewed, revised and finalized the sketches. Apart from that, all content, results and conclusions were developed independently; no AI-generated text was adopted. The tool was used in line with the principles of *responsible AI use* (transparency, verifiability, human oversight). Sources found through research were checked for accuracy independently. All content was reviewed, validated and finally approved by the lab members (cf. FH Salzburg AI guideline *KI-Richtlinie*, version 1.0, ch. 1.3 and 2.1.2).